**Compiled by Maximus Pung Jun**
**H2 Mathematics (9758)**
**2026**

## 2026 H2 Math Vectors I Homework 1: Solutions

Introduction to Vectors in Three Dimensions. Original worked solutions. Total: 50 marks. Give exact answers unless otherwise stated.

**1** The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a}=2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+4\mathbf{j}$.

**(a)** Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a}+2\mathbf{b}$. [2]

**(b)** Find $\mathbf{c}$ such that $2\mathbf{a}-3\mathbf{b}+\mathbf{c}=\mathbf{0}$. [2]

**(c)** Points $P$ and $Q$ have position vectors $\mathbf{a}+2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. [1]

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}, \quad \mathbf{b}=\begin{pmatrix}-1\\4\\0\end{pmatrix},$<br>$\mathbf{a}+2\mathbf{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-1\\4\\0\end{pmatrix}$<br>$=\begin{pmatrix}0\\7\\3\end{pmatrix}.$ | |
| **(b)** **[2]** | $\mathbf{c}=3\mathbf{b}-2\mathbf{a}=\begin{pmatrix}-3\\12\\0\end{pmatrix}-\begin{pmatrix}4\\-2\\6\end{pmatrix}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}.$ | |
| **(c)** **[1]** | $\overrightarrow{PQ}=\overrightarrow{OQ}-\overrightarrow{OP}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}-\begin{pmatrix}0\\7\\3\end{pmatrix}=\begin{pmatrix}-7\\7\\-9\end{pmatrix}.$ | |

**2** The points $A$ and $B$ have coordinates $(1, -2, 2)$ and $(3, 1, -4)$ respectively.

**(a)** Find a unit vector that is parallel to $AB$. [2]

**(b)** Find a vector of length 21 parallel to $AB$. [2]

**(c)** A point $P$ is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of $P$ and state the order of the three points along their common straight path. [2]

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$, $\quad |\overrightarrow{AB}| = \sqrt{4+9+36} = 7.$<br>One such unit vector is $\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$. Its negative is also a valid answer. | |
| **(b)** **[2]** | One such vector is $21\left(\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}\right) = \begin{pmatrix} 6 \\ 9 \\ -18 \end{pmatrix}$. Its negative is also a valid answer. | |
| **(c)** **[2]** | $\overrightarrow{AP} = 14\left(\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 6 \\ -12 \end{pmatrix}$, $\quad \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \begin{pmatrix} 5 \\ 4 \\ -10 \end{pmatrix}$.<br>Thus $P(5, 4, -10)$. Since $\overrightarrow{AP} = 2\overrightarrow{AB}$, the order is $A, B, P$. | |

**3** The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where $k$ and $m$ are real constants.

**(a)** Find $k$ and $m$. [3]

**(b)** State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. [1]

**(c)** Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. [2]

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[3]** | Since the vectors are parallel, $\mathbf{p}=h\mathbf{q}$ for some non-zero real $h$.<br>Comparing first components gives $6=2h$, so $h=3$.<br>$k=3(-4)=-12, \quad -3=3m \implies k=-12,\ m=-1.$ | |
| **(b)** **[1]** | Since $\mathbf{p}=3\mathbf{q}$, they have the same direction and $\lvert\mathbf{p}\rvert:\lvert\mathbf{q}\rvert=3:1$. | |
| **(c)** **[2]** | $\lvert\mathbf{q}\rvert=\sqrt{2^2+(-4)^2+(-1)^2}=\sqrt{21}$. Both directions must be included, so all such vectors are<br>$\pm\begin{pmatrix}2\\-4\\-1\end{pmatrix}$. | |

**4** The points $A$, $B$ and $C$ have coordinates $(1, 2, -1)$, $(3, 1, 2)$ and $(7, -1, 8)$ respectively.

**(a)** Show that $A$, $B$ and $C$ are collinear, and find $AB : BC$. [3]

**(b)** The point $R(u, -2, v)$ lies on the same straight path as $A$, $B$ and $C$. Find $u$ and $v$. [3]

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\overrightarrow{AB} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} = 2\overrightarrow{AB}$.<br>Thus $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel and have the common point $B$, so $A, B, C$ are collinear. Taking magnitudes gives $AB : BC = 1 : 2$. | |
| **(b)** **[3]** | Collinearity gives $\overrightarrow{AR} = s\overrightarrow{AB}$ for some real $s$, so<br>$\begin{pmatrix} u-1 \\ -4 \\ v+1 \end{pmatrix} = s \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$.<br>The second components give $s = 4$. Hence $u - 1 = 8$ and $v + 1 = 12$, giving $u = 9$, $v = 11$. | |

**5** The points $A$ and $B$ have coordinates $(2,-1,3)$ and $(-4,5,0)$ respectively. A point $M$ divides $AB$ internally in the ratio $AM:MB=2:1$. A point $N$ lies on $AB$ produced beyond $B$ and satisfies $AN:NB=5:2$.

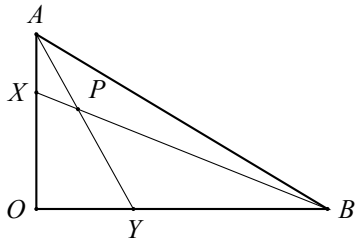
**(a)** Find the position vector of $M$. [2]

**(b)** Find the position vector of $N$. [3]

**(c)** State the order of the four points $A$, $B$, $M$ and $N$, and find $MB:BN$. [3]

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | By the ratio theorem, $\overrightarrow{OM}=\dfrac{\overrightarrow{OA}+2\overrightarrow{OB}}{3}=\dfrac{1}{3}\left[\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-4\\5\\0\end{pmatrix}\right]=\begin{pmatrix}-2\\3\\1\end{pmatrix}$. | |
| **(b)** **[3]** | Since $AN=AB+BN$ and $AN:BN=5:2$, $AB:BN=3:2$. Hence $\overrightarrow{BN}=\frac{2}{3}\overrightarrow{AB}$, and $\overrightarrow{ON}=\overrightarrow{OB}+\dfrac{2}{3}(\overrightarrow{OB}-\overrightarrow{OA})=\dfrac{5\overrightarrow{OB}-2\overrightarrow{OA}}{3}=\begin{pmatrix}-8\\9\\-2\end{pmatrix}$. | |
| **(c)** **[3]** | The order is $A, M, B, N$. Since $MB=\frac{1}{3}AB$ and $BN=\frac{2}{3}AB$, $MB:BN=1:2$. | |

**6** Relative to the origin $O$, the points $A$ and $B$ have non-zero, non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point $X$ lies on $OA$ with $OX:XA=2:1$, and the point $Y$ lies on $OB$ with $OY:YB=1:2$. The line segments $AY$ and $BX$ intersect at $P$.




**(a)** By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [6]

**(b)** Find $AP:PY$ and $BP:PX$. [2]

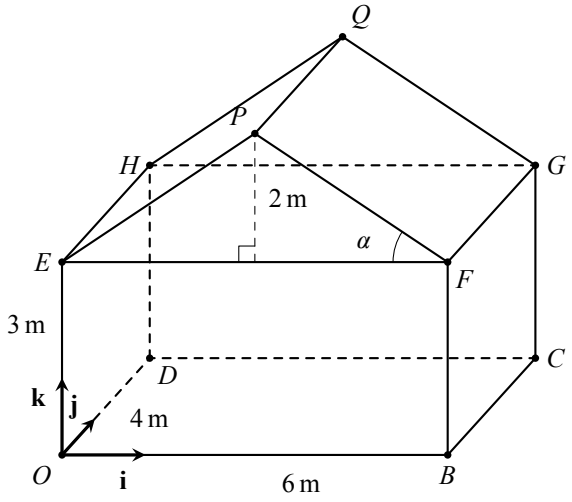
**(c)** The point $Z$ divides $AB$ internally with $AZ:ZB=1:4$. Show that $O, P, Z$ are collinear and find $OP:PZ$. [2]

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[6]** | First, $\overrightarrow{OX}=\frac{2}{3}\mathbf{a}$ and $\overrightarrow{OY}=\frac{1}{3}\mathbf{b}$. Let $AP:PY=s:(1-s)$ and $BP:PX=t:(1-t)$, where $0<s,t<1$. By the ratio theorem,<br>$\overrightarrow{OP}=(1-s)\overrightarrow{OA}+s\overrightarrow{OY}=(1-s)\mathbf{a}+\frac{s}{3}\mathbf{b}$,<br>$\overrightarrow{OP}=(1-t)\overrightarrow{OB}+t\overrightarrow{OX}=\frac{2t}{3}\mathbf{a}+(1-t)\mathbf{b}$.<br>Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, comparing coefficients gives<br>$1-s=\frac{2t}{3}, \qquad \frac{s}{3}=1-t$.<br>The second equation gives $s=3-3t$. Substituting into the first,<br>$-2+3t=\frac{2t}{3} \implies 7t=6 \implies t=\frac{6}{7}, \quad s=\frac{3}{7}$.<br>Consequently,<br>$\overrightarrow{OP}=\frac{4}{7}\mathbf{a}+\frac{1}{7}\mathbf{b}$. | |
| **(b)** **[2]** | $AP:PY=3:4, \qquad BP:PX=6:1$. | |
| **(c)** **[2]** | By the ratio theorem,<br>$\overrightarrow{OZ}=\frac{4\mathbf{a}+\mathbf{b}}{5}, \qquad \overrightarrow{OP}=\frac{4\mathbf{a}+\mathbf{b}}{7}=\frac{5}{7}\overrightarrow{OZ}$.<br>The two vectors are parallel and have the common point $O$, so $O, P, Z$ are collinear. Since $0<\frac{5}{7}<1$, $P$ is between $O$ and $Z$; hence $OP:PZ=5:2$. | |

**7** A solid house consists of a cuboid *OBCDEFGH* and a triangular prism roof *EFPHGQ*, as shown. The base *OBCD* is horizontal. The vertices *E*, *F*, *G* and *H* are directly above *O*, *B*, *C* and *D* respectively.

The front-left corner *O* is the origin. The positive **i**, **j** and **k** directions are along *OB*, *OD* and *OE* respectively, and each unit represents 1 metre. It is given that $OB = 6$ m, $OD = 4$ m and $OE = 3$ m.

The ridge *PQ* is parallel to *OD*. The front gable *EFP* and back gable *HGQ* are congruent isosceles triangles, with *P* vertically above the midpoint of *EF* and *Q* vertically above the midpoint of *HG*. Each sloping roof face makes an angle $\alpha$ with the horizontal, where $\tan\alpha = \frac{2}{3}$; the vertical roof rise is 2 m.



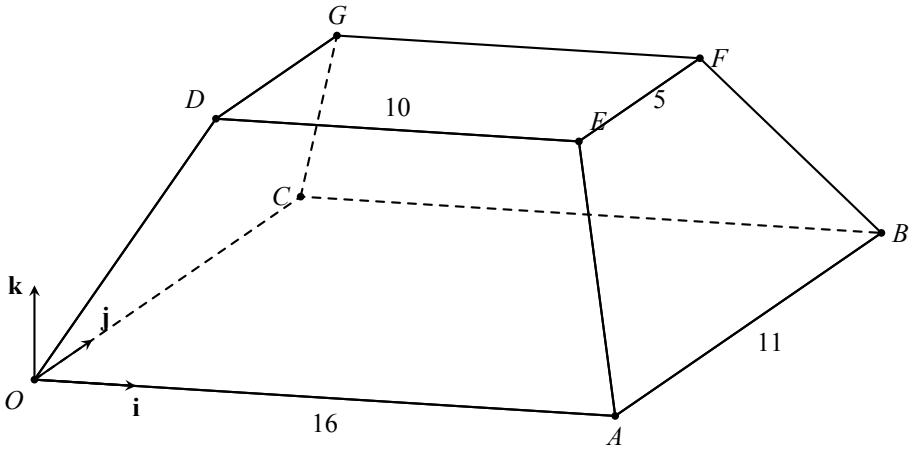

All lengths are in metres. The diagram is not drawn to scale.

Find the coordinates of all the vertices of the house $(O, B, C, D, \ldots)$. [6]

| Solution | | Comments |
|---|---|---|
| **[6]** | The positive coordinate directions are along $OB$, $OD$ and $OE$. Hence the four base vertices have $z=0$, the front vertices have $y=0$, and the back vertices have $y=4$.<br>The walls are 3 m high, so $E, F, G, H$ all have $z=3$. The roof is symmetric across the middle of its 6 m width, so the ridge has $x=\frac{6}{2}=3$.<br>Writing $h$ for the vertical roof rise,<br>$\tan\alpha=\dfrac{h}{6/2}=\dfrac{2}{3} \quad\Longrightarrow\quad h=2\,\text{m}.$<br>This agrees with the labelled rise. The ridge is at height $3+2=5$ m. Therefore all ten vertices are<br>$O=(0,0,0),\quad B=(6,0,0),$<br>$C=(6,4,0),\quad D=(0,4,0),$<br>$E=(0,0,3),\quad F=(6,0,3),$<br>$G=(6,4,3),\quad H=(0,4,3),$<br>$P=(3,0,5),\quad Q=(3,4,5).$ | |

**8** The diagram shows a container with a horizontal rectangular base $OABC$, where $OA = 16$ cm and $AB = 11$ cm. The top of the container $DEFG$ is also a horizontal rectangle, where $DE = 10$ cm and $EF = 5$ cm. The four sloping faces (for example $OAED$) of the container, each a trapezium, are inclined at the same angle to the horizontal, and the vertical distance between the base and the top is 6 cm.

The point $O$ is taken as the origin $(0, 0, 0)$, and perpendicular unit vectors **i**, **j**, **k** are such that **i** is parallel to $OA$, **j** is parallel to $OC$, and **k** is vertical.




All lengths are in centimetres. The diagram is not drawn to scale.

Find the coordinates of $D$. [3]

| Solution | | Comments |
|---|---|---|
| **[3]** | Since the four sloping faces are inclined at the same angle to the horizontal, the top rectangle $DEFG$ is centred above the base rectangle $OABC$.<br>The horizontal inset from each base edge to the corresponding top edge is<br>$\frac{OA-DE}{2}=\frac{16-10}{2}=3, \qquad \frac{AB-EF}{2}=\frac{11-5}{2}=3.$<br>The top lies 6 cm above the base, so the $z$-coordinate of every top vertex is 6. Taking $D$ as the front-left vertex of the top,<br>$D=(3, 3, 6)$. | |

**END OF SOLUTIONS**